Level-Crossing Density as a Mesh-Free High-Frequency Auxiliary Loss for Implicit Neural Representations

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Abstract

Coordinate-MLP implicit neural representations (INRs) exhibit *spectral bias*: they fit low-frequency structure quickly and high-frequency detail slowly or not at all. Existing remedies act in the frequency domain (e.g., the Focal Frequency Loss), on activations and encodings (SIREN, Fourier features, FINER), or through coarse-to-fine curricula—and the loss-based remedies presuppose a regular sampling grid on which a discrete Fourier transform is available. We revisit a classical object from random-field theory, the *Rice level-crossing density*, and turn it into a differentiable training objective. By the co-area formula, the density of level crossings of a field is a direct functional of its gradient magnitude concentrated on level sets, and for stationary processes it is monotone in the ratio of spectral moments $\sqrt{\lambda_2/\lambda_0}$—i.e., it is a spatial-domain, closed-form proxy for high-frequency content. We construct a smoothed Monte-Carlo estimator of the crossing density that is evaluated pointwise at arbitrary sample locations, requires no grid, no FFT, and no mesh, and match it to the empirical crossing density of the supervision signal across a set of levels. We validate the estimator against exact crossing counts and the Rice formula, then evaluate the loss under strict information parity against the Focal Frequency Loss, normalized Sobolev supervision, and MSE-only training, on regular grids and on scattered non-uniform samples, and we report the outcome without embellishment. Where supervision is scarce and irregular, *every* auxiliary spectral loss is transformative (+2.3–3.0 dB over MSE-only on PE-MLP, +1.4–1.8 dB on SIREN); on an edge-dominated natural image the crossing-density loss matches the alternatives—within 0.2–0.5 dB of FFL on PSNR, matching or exceeding it on SSIM—without beating them, while on a statistically homogeneous texture, the regime closest to the stationary fields of the Rice theory, it overtakes FFL by 0.6 dB. Its further distinctions are structural: it is the only loss requiring neither a sampling grid nor trustworthy pointwise gradient targets, and an oracle-gradient diagnostic shows it is uniquely insensitive to gradient-target quality. On dense regular grids, where auxiliary drives are unnecessary, finite-difference gradient targets actively hurt—a caution that applies equally to Sobolev training. We release code, raw results, and all experiment scripts.

1 Introduction

Implicit neural representations (INRs) parameterize a signal as a neural network $f_\theta : \mathbb{R}^d \to \mathbb{R}^m$ mapping coordinates to values, and have become a standard representation for images, signed distance fields, and radiance fields [16, 18, 24, 28]. Their best-known failure mode is *spectral bias* [3, 20]: trained with a plain reconstruction loss, coordinate MLPs learn low frequencies first and high frequencies slowly, blurring fine detail. The community's responses fall into three families: (i) *architectural* fixes—Fourier feature encodings [25], periodic activations [24], their variable-periodic [14], wavelet [23], and Gaussian [21] relatives, multiplicative filter networks [6], band-limited architectures [13], and hash grids [17]; (ii) *loss-based* fixes—frequency-domain objectives such as the Focal Frequency Loss (FFL) [9] and gradient-domain (Sobolev) supervision [4, 30]; and (iii) *curricula* that schedule frequency content coarse-to-fine [8, 29].

The loss-based family has a structural blind spot. A discrete Fourier transform is defined on a regular grid. When supervision arrives as scattered samples—point clouds for neural SDFs, non-uniformly sampled sensor data, adaptively sampled renderings—a frequency-domain loss must first *resample* the scattered data onto a grid, and the interpolation step smears exactly the high-frequency content the loss is meant to supervise. Gradient-matching losses are pointwise and thus mesh-free, but they supervise the gradient *at each point*, so their targets inherit, pointwise and at full strength, whatever noise the gradient-estimation step introduces.

This paper proposes a third kind of spatial-domain objective, borrowed from random-field theory. The *Rice formula* [10, 22] states that for a stationary Gaussian process the expected number of crossings of level u per unit length is

$$\mu(u) \;=\; \frac{1}{\pi}\sqrt{\frac{\lambda_2}{\lambda_0}}\; e^{-u^2/(2\lambda_0)}, \qquad \lambda_k = \int \omega^k \, dS(\omega), \tag{1}$$

where S is the spectral measure. The crossing rate is monotone in $\sqrt{\lambda_2/\lambda_0}$, the RMS angular frequency: *how often a field crosses a level is a direct, closed-form functional of how much high-frequency energy it carries.* This link between zero-crossing rates and spectral content is classical in signal processing [11, 15]; our contribution is not the link itself but its *instantiation as a differentiable training objective for INRs*, and an empirical study of where such an objective helps.

Concretely, we estimate the crossing density of the INR's output field with a smoothed Monte-Carlo form of the Kac–Rice integrand,

$$\hat{c}_\varepsilon(u) \;=\; \frac{1}{N}\sum_{i=1}^{N} \delta_\varepsilon\big(f_\theta(x_i) - u\big)\,\big\|\nabla_x f_\theta(x_i)\big\|, \tag{2}$$

where the x_i are the (arbitrary, possibly scattered) training coordinates, δ_ε is a Gaussian bump, and $\nabla_x f_\theta$ is exact via automatic differentiation. By the co-area formula, (2) estimates crossings per unit length in 1D, level-set length per unit area in 2D, and level-set area per unit volume in 3D. The auxiliary loss matches $\hat{c}_\varepsilon(u)$ to the *empirical* crossing density of the ground truth at a small set of levels. Everything is pointwise: no grid, no FFT, no mesh, no stationarity or Gaussianity assumption.

Because a crossing-density profile is a *distributional* statistic—an average over the whole batch—it discards phase information: it says how much level-set geometry the field should have, not where. It is therefore an auxiliary drive alongside a reconstruction loss (which anchors locations), not a stand-alone objective, and its distributional character is exactly what we expect to make it robust on scattered domains: pointwise errors in derived gradient targets average out in (2) instead of being enforced point by point.

Claims. We make three claims, each tested in Section 5. (1) The estimator (2) is an accurate, differentiable crossing-density estimator (validated against exact counts and against (1)). (2) As an auxiliary loss it is a *working* high-frequency drive: wherever supervision is scarce it recovers most of the gap between MSE-only training and the information ceiling set by the samples, on two different backbones. (3) Against the strongest loss-based alternatives under strict information parity, it achieves PSNR parity on edge-dominated natural-image content (with SSIM matching or exceeding FFL's and unique robustness to gradient-target quality) and overtakes FFL only on statistically homogeneous texture—the regime the Rice theory actually describes. We consider the disciplined measurement of (3), including its negative component, a contribution: the hypothesis that crossing statistics would broadly *beat* resampled frequency losses off-grid was plausible, cheap to state, and is here tested and bounded.

What we do not claim. Coarse-to-fine frequency curricula exist [8, 29] and we do not claim one. The crossing-rate–frequency link is textbook [11]; we claim only the differentiable-loss instantiation and its evaluation. On regular grids we expect, and report, rough parity rather than dominance.

2 Related work

Spectral bias and architectural remedies. Spectral bias was characterized by Rahaman et al. [20], with Basri et al. [3] extending the analysis to non-uniform input densities—directly relevant to our scattered-sampling setting, since their theory predicts that convergence at a given frequency slows further in sparsely sampled regions. Fourier feature encodings [25] and periodic activations [24] move the representable band up; FINER [14] makes the supported band tunable via variable-periodic activations; WIRE [23], Gaussian activations [21], multiplicative filter networks [6], BACON [13], and hash encodings [17] are alternative routes. All of these change the *architecture*; our proposal is orthogonal and combines with them (Section 5.4).

Loss-based remedies. The Focal Frequency Loss [9] compares DFT spectra of prediction and target with an adaptive per-frequency weight and is the standard frequency-domain objective; it requires a regular grid. Sobolev training supervises derivatives [4], applied to INRs with finite-difference image derivatives by Yuan et al. [30]; it is mesh-free but pointwise. Eikonal-type losses in shape representation [7] likewise constrain $\|\nabla f\|$ pointwise (to unit norm for SDFs); our loss instead constrains an integral functional of $\|\nabla f\|$ restricted to level sets. Frequency curricula [8, 29] schedule the encoding rather than add an objective.

Level crossings and the Kac–Rice formula. Crossing counts of random processes go back to Kac [10] and Rice [22]; modern treatments include Adler and Taylor [1], Azaïs and Wschebor [2], and Longuet-Higgins [15] developed the 2D (level-set length) theory for random surfaces. Zero-crossing rates as frequency features are classical in signal processing and time-series discrimination [11]. To our knowledge—including searches over arXiv abstracts and the recent INR literature—level-crossing densities have not previously been used as a training objective for neural fields; the Kac–Rice machinery appears in machine learning mainly for counting critical points of loss landscapes, a different use entirely.

3 Method

3.1 Background: crossings, spectra, and the co-area identity

Let $f : \Omega \subset \mathbb{R}^d \to \mathbb{R}$ be Lipschitz. For $d = 1$ and a level u, let $N_u(f; \Omega)$ be the number of solutions of $f(x) = u$ in Ω. For stationary Gaussian processes, Rice's formula (1) gives the expected crossing rate as a ratio of spectral moments: λ_0 is the signal variance and λ_2 the derivative variance, so $\sqrt{\lambda_2/\lambda_0}$ is an RMS frequency. A field that lacks high-frequency energy crosses every level less often (Figure 2, right).

Two properties make the crossing density usable beyond the Gaussian setting. First, the *Kac–Rice integrand*: under mild conditions [2],

$$\mathbb{E}\, N_u \;=\; \int_\Omega \mathbb{E}\big[\, |f'(x)| \,\big|\, f(x) = u \,\big]\, p_{f(x)}(u)\, dx, \tag{3}$$

which requires no stationarity. Second, and central for us, the deterministic *co-area formula* [5]: for Lipschitz f and integrable g,

$$\int_\Omega g\big(f(x)\big)\, \|\nabla f(x)\|\, dx \;=\; \int_{\mathbb{R}} g(u)\, \mathcal{H}^{d-1}\big(f^{-1}(u) \cap \Omega\big)\, du, \tag{4}$$

where $\mathcal{H}^{d-1}$ is the $(d{-}1)$-dimensional Hausdorff measure. Taking $g = \delta_\varepsilon(\cdot - u)$, a Gaussian kernel of bandwidth ε, the left-hand side of (4) divided by $|\Omega|$ becomes an ε-smoothed *level-set density*: crossings per unit length (d=1), level-set length per unit area (d=2), level-set area per unit volume (d=3). No probabilistic assumptions enter: (4) holds for the single deterministic field being fitted.

3.2 A differentiable Monte-Carlo estimator

Sampling coordinates $x_1, \ldots, x_N$ uniformly in Ω (or using the given training coordinates) yields the estimator of (2): $\hat{c}_\varepsilon(u) = \frac{1}{N} \sum_i \delta_\varepsilon(f_\theta(x_i) - u)\, \|\nabla_x f_\theta(x_i)\|$ with $\delta_\varepsilon(z) = (2\pi\varepsilon^2)^{-1/2} e^{-z^2/2\varepsilon^2}$. It is unbiased for the ε-smoothed level-set density and inherits the usual kernel trade-off: $O(\varepsilon^2)$ smoothing bias across levels versus $O(1/(N\varepsilon))$ variance. Three implementation points matter in practice.

Gradients are free and exact. $\nabla_x f_\theta$ comes from automatic differentiation at arbitrary points; no finite differences, no grid. Backpropagating the loss through $\|\nabla_x f_\theta\|$ is a double-backward operation, supported by all standard frameworks [19].

Levels at target quantiles. We place L levels $u_1, \ldots, u_L$ at uniformly spaced quantiles of the ground-truth values in the batch (between the 2nd and 98th percentile), which keeps every level populated regardless of the value histogram.

Same-batch targets. The target profile $c^{\mathrm{gt}}(u_j)$ is the same estimator applied to the ground-truth values and gradient norms *at the same batch points*. Estimator and target then share sampling noise, which partially cancels in the difference—the loss compares two estimates of the same functional under the same measure, so even under non-uniform sampling both sides are biased identically toward densely sampled regions.

3.3 The Kac–Rice loss

The auxiliary loss matches the crossing-density profile at the L levels, normalized to be scale-free:

$$\mathcal{L}_{\mathrm{KR}}(\theta) \;=\; \frac{1}{L}\sum_{j=1}^{L} \frac{\big(\hat{c}_\varepsilon(u_j) - c^{\mathrm{gt}}(u_j)\big)^2}{\big(c^{\mathrm{gt}}(u_j) + \bar{c}^{\mathrm{gt}}\big)^2}, \qquad \bar{c}^{\mathrm{gt}} = \tfrac{1}{L}\textstyle\sum_j c^{\mathrm{gt}}(u_j), \tag{5}$$

and the total objective is $\mathcal{L} = \frac{1}{N}\sum_i (f_\theta(x_i) - y_i)^2 + \beta\,\mathcal{L}_{\mathrm{KR}}$. The normalization in (5) is not cosmetic: crossing densities grow linearly with frequency content, so the *squared* error grows quadratically, and an unnormalized version couples the effective weight β to the signal's bandwidth. In early experiments the unnormalized loss with a weight tuned for one signal destabilized training on another; the relative form transfers across signals with a single β.[1]

Defaults throughout: $L = 16$ levels, $\varepsilon = 0.15\,\sigma_y$ where σ_y is the standard deviation of the batch ground-truth values, and $\beta = 0.05$; all are ablated in Section 5.5.

What the loss constrains. $\mathcal{L}_{\mathrm{KR}}$ is a functional of the *marginal* level-set geometry: it pushes the field to have the right amount of level-crossing structure at each height, leaving *where* that structure goes to the reconstruction term. It cannot, alone, place an edge; it can insist that the total edge length at each gray level be right. This is weaker than a Fourier loss on a grid (which constrains amplitude at every frequency) but requires strictly less of the sampling process, and it is robust to pointwise target noise by construction: a zero-mean error of variance σ^2 on individual gradient-norm targets perturbs each $c^{\mathrm{gt}}(u_j)$ only at $O(\sigma/\sqrt{N_j})$, with N_j the effective number of points near level j, whereas a pointwise gradient-matching loss absorbs that noise at full variance.

Cost. Relative to plain MSE, the additional cost is one backward pass for $\nabla_x f_\theta$ (shared with any gradient-based loss) plus an $O(NL)$ kernel evaluation, and double-backprop at optimization time. In a back-to-back measurement (PE-MLP, 16,384 points, CPU; `bench_timing.py` in the released code, raw output in the result files), one training iteration took 223 ms with MSE only, 473 ms with the Kac–Rice term (2.12×), and 479 ms with Sobolev (2.14×)—the two gradient-based losses share the dominant double-backward cost.

4 Estimator validation

Before using (2) inside a loss we verify it in isolation (Figure 1). On deterministic multisine signals the estimator matches analytic crossing counts to within Monte-Carlo error (e.g., $\sin(\pi k x)$ on $[-1, 1]$ yields measured density $k \pm 2\%$ at $N = 2 \times 10^5$, and the 2D co-area version recovers the level-set length density of oriented gratings equally well). On an approximately Gaussian random field (60 random-phase sinusoids), the estimator agrees with direct sign-change counting on a dense grid within 5% across levels $u \in [-2.5\sigma, 2.5\sigma]$, and with the Rice prediction (1) evaluated at empirical spectral moments within 10%. These tests ship as unit tests with the code.

5 Experiments

5.1 Protocol

Backbones. The primary backbone for loss comparisons is a ReLU MLP on Fourier positional encodings (PE-MLP; 3 hidden layers, width 256), the classic spectrally biased INR [25]. SIREN [24] and FINER [14] appear as architecture baselines under plain MSE, and we test composability by adding our loss to SIREN.

Baselines. (i) MSE only; (ii) MSE + Focal Frequency Loss [9]—our implementation is numerically verified against the official package; (iii) MSE + normalized Sobolev (H^1 gradient matching) [4, 30].

[1] The same reasoning applies to the Sobolev baseline, which we normalize by the mean squared ground-truth gradient norm; without this, its useful weight range also varies by orders of magnitude across signals.

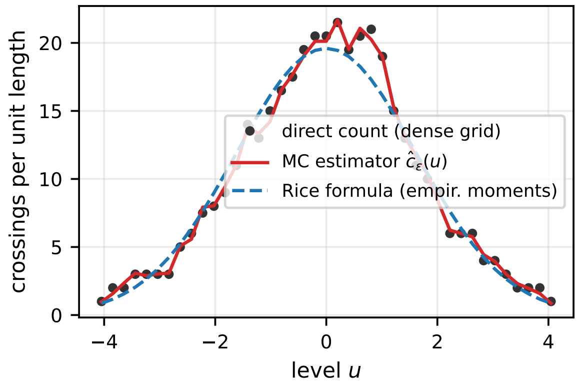

Figure 1: Estimator validation. Smoothed Monte-Carlo estimator $\hat{c}_\varepsilon(u)$ (red, N=4 × 10^5 points, ε=0.05), exact crossing counts from dense-grid sign changes (dots), and the Rice formula at empirical spectral moments (dashed), for an approximately Gaussian random field (60 random-phase sinusoids). The estimator tracks the exact counts across all levels, including the non-Gaussian wiggles that the Rice formula idealizes away.

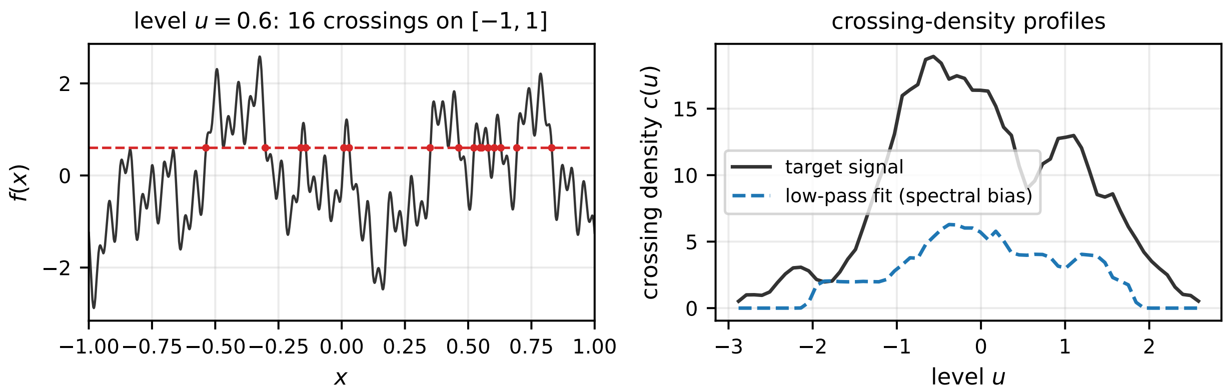

Figure 2: Left: a multisine signal, one level u, and its crossings; the crossing density at level u is the number of such intersections per unit length. **Right:** crossing-density profiles $c(u)$ of the full signal versus its low-pass approximation—the profile collapses when high frequencies are missing, which is the signal the loss exploits.

Tasks. *Regular grids* (sanity check): a five-tone multisine in 1D (frequencies $2, 5, 11, 23, 47$ half-cycles per unit) fitted from 1,024 grid points, and the `camera` image at 128^2. *Scattered domains* (the decisive test): the 256^2 `camera` image supervised only at N=8,192 scattered locations (12.5% of pixels) drawn from three densities—`uniform` (homogeneous), `blobs` (75% of samples in five Gaussian clusters), and `ramp` (linear density gradient)—plus a 1D variant and a synthetic multi-scale texture. On scattered tasks *all* losses receive exactly the same information: the scattered pairs $\{(x_i, y_i)\}$. FFL's grid target is obtained by Delaunay-based linear interpolation (SciPy `griddata` [26]) and the loss is applied on random 128^2 crops of that grid each iteration (patch-based FFL, a mode the official implementation supports); Sobolev and Kac–Rice gradient targets are central differences *of that same interpolant* sampled at the training points. Oracle-gradient variants (true image gradients; not legal supervision) isolate how much each gradient-based method loses to interpolation noise.

Metrics. PSNR on a dense evaluation grid; high-frequency PSNR (HF-PSNR: PSNR of the residual after removing a Gaussian low-pass, $\sigma = 4\,\text{px}$); SSIM [27]; and relative spectral error in eight radial frequency bands. All results are mean ± s.d. over three seeds (model initialization and, on scattered tasks, the sampling pattern). Adam [12] with cosine decay; full hyperparameters in Appendix A.

Table 1: Regular-grid fitting (sanity check; mean ± s.d. over 3 seeds). 1D: parity across all losses. 2D: dense supervision needs no auxiliary drive—MSE-only is best among PE-MLP losses, and architecture fixes (SIREN, FINER) dominate the entire loss family on-grid.

	1D multisine	2D camera 128^2		
Config	PSNR	PSNR	HF-PSNR	SSIM
MSE only	70.64 ± 0.11	40.52 ± 1.10	42.46 ± 1.10	0.97 ± 0.00
+ FFL	69.72 ± 0.23	37.85 ± 0.52	39.80 ± 0.52	0.95 ± 0.00
+ Sobolev	70.41 ± 0.22	29.30 ± 0.12	31.26 ± 0.12	0.90 ± 0.00
+ Kac-Rice (ours)	70.28 ± 0.30	30.80 ± 1.00	32.79 ± 1.00	0.89 ± 0.02
SIREN (MSE)	93.77 ± 1.09	62.41 ± 0.33	64.33 ± 0.33	1.00 ± 0.00
FINER (MSE)	73.57 ± 7.68	139.72 ± 3.09	141.56 ± 3.04	1.00 ± 0.00

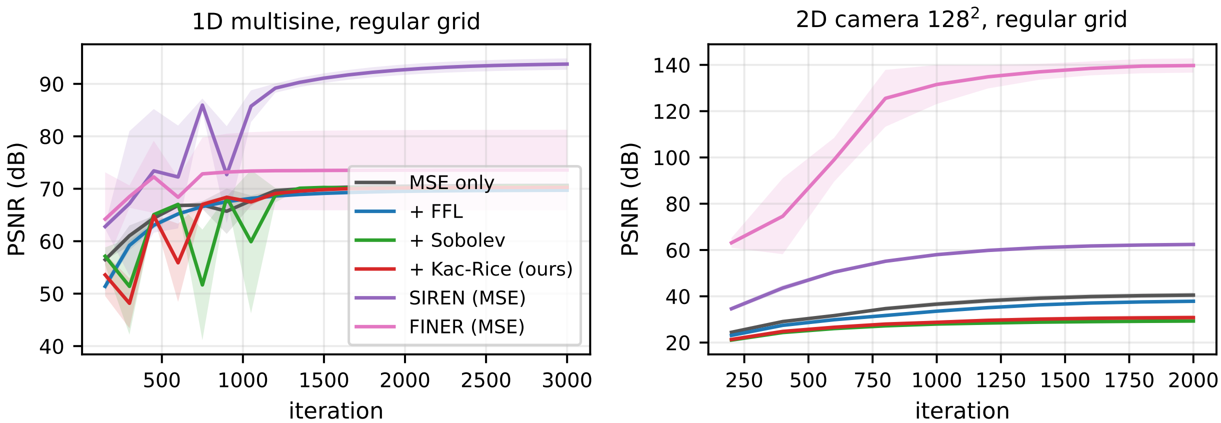

Figure 3: Regular-grid convergence. PSNR versus iteration (mean over 3 seeds; shaded ± s.d.). Left: 1D multisine—all PE-MLP losses converge to parity. Right: 128^2 image—with dense supervision, MSE-only is best and the finite-difference gradient targets of Sobolev/Kac–Rice actively cost fidelity (see text).

5.2 Regular grids: parity, as expected

1D: parity. On the grid-sampled multisine, all four PE-MLP losses converge to statistically indistinguishable quality (MSE 70.6 ± 0.1 dB, FFL 69.7 ± 0.2, Sobolev 70.4 ± 0.2, Kac–Rice 70.3 ± 0.3): with dense regular supervision, the reconstruction term alone eventually recovers all five tones, and no auxiliary drive changes the endpoint. SIREN, whose periodic activations match this signal family, reaches 93.8 ± 1.1 dB; FINER reaches 73.6 ± 7.7 dB with high seed variance. This is the expected sanity-check outcome: on-grid, auxiliary spectral losses neither help nor hurt materially in 1D.

2D: dense supervision does not need—and is hurt by—auxiliary drives. On the 128^2 image (Table 1), MSE-only training is the *best* configuration (40.5 ± 1.1 dB); FFL costs 2.7 dB, and the two gradient-based losses cost $\sim$10 dB. The mechanism for the latter is instructive: the gradient targets are central differences of the discrete image, which are a low-pass-filtered version of the true local differences the MSE term is driving toward. At high fidelity the two objectives conflict, and a fixed β biases the optimum away from exact pixel reconstruction. (FFL's target is the exact pixel grid, so it remains consistent with MSE and costs little.) Yuan et al. [30] similarly report that Sobolev training for INRs requires care in how image derivatives are approximated. The architecture baselines put the loss family in perspective: on-grid, SIREN reaches 62.4 ± 0.3 dB and FINER 139.7 ± 3.1 dB (a numerically exact fit)—dense grid fitting is simply solved by activation engineering, and no auxiliary loss on a PE-MLP competes there. The practical reading, consistent with the few-shot literature [29]: auxiliary spectral objectives are tools for *underdetermined* regimes, not for dense overfitting, and we evaluate them where they are meant to operate—Section 5.3.

Table 2: Scattered-domain fitting: the decisive comparison (256^2 `camera`, N=8,192 samples, three sampling densities; mean ± s.d. over 3 seeds). All losses receive only the scattered samples. The `griddata` row is the plain linear-interpolation reference computed from the same samples.

	blobs		ramp		uniform	
Config	PSNR	HF-PSNR	PSNR	HF-PSNR	PSNR	HF-PSNR
griddata interp. (input)	21.84 ± 0.10	25.27 ± 0.01	22.85 ± 0.10	25.70 ± 0.06	23.11 ± 0.10	25.84 ± 0.08
MSE only	18.88 ± 0.12	22.76 ± 0.02	19.73 ± 0.09	22.89 ± 0.07	20.18 ± 0.07	23.19 ± 0.05
+ FFL (interp.)	21.62 ± 0.12	25.01 ± 0.05	22.48 ± 0.12	25.31 ± 0.09	22.73 ± 0.11	25.46 ± 0.09
+ Sobolev (est. grad)	21.59 ± 0.15	25.07 ± 0.09	22.74 ± 0.07	25.59 ± 0.05	23.02 ± 0.08	25.75 ± 0.05
+ Kac-Rice (est. grad, ours)	21.14 ± 0.17	24.71 ± 0.07	22.31 ± 0.15	25.25 ± 0.12	22.53 ± 0.15	25.36 ± 0.13
+ Sobolev (oracle grad)	21.80 ± 0.21	25.36 ± 0.13	–	–	–	–
+ Kac-Rice (oracle grad)	20.86 ± 0.20	24.50 ± 0.08	–	–	–	–
SIREN (MSE)	20.10 ± 0.20	23.97 ± 0.15	–	–	–	–
SIREN + FFL (interp.)	21.92 ± 0.11	25.38 ± 0.03	–	–	–	–
SIREN + Sobolev (est. grad)	21.50 ± 0.14	25.06 ± 0.10	–	–	–	–
SIREN + Kac-Rice (ours)	21.59 ± 0.10	25.14 ± 0.05	–	–	–	–

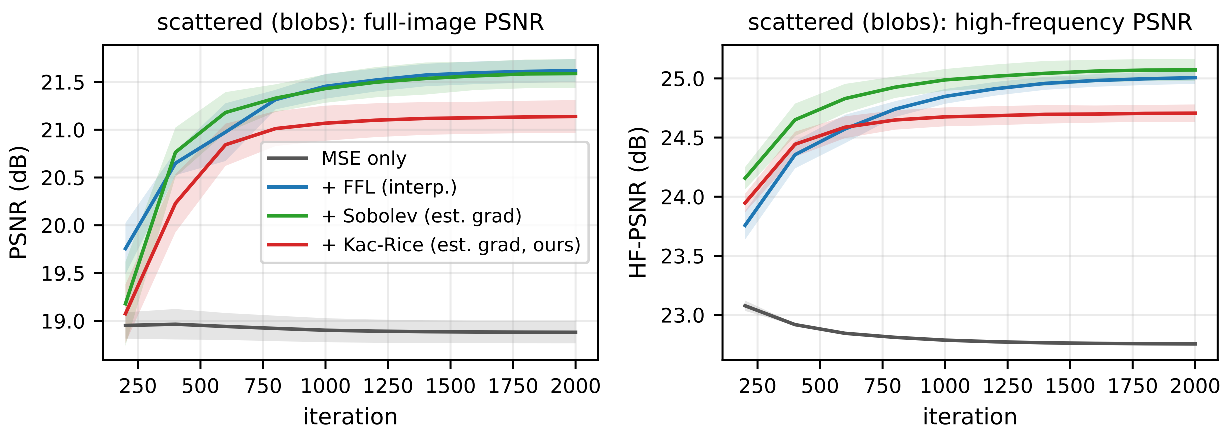

Figure 4: Scattered-domain convergence (`blobs` sampling; mean over 3 seeds, shaded ± s.d.). Full-image PSNR (left) and high-frequency PSNR (right). MSE-only stalls immediately; all three auxiliary losses climb 2.3–2.7 dB above it. Kac–Rice leads on HF-PSNR early (its target does not need to warm up any grid structure) before Sobolev and FFL edge ahead after ∼700 iterations.

5.3 Scattered non-uniform domains: the decisive test

All auxiliary losses matter here; Kac–Rice ties FFL but does not beat it. Table 2 is the decisive comparison, and we report it as measured. First, the regime change from Section 5.2 is dramatic: with scattered supervision, *every* auxiliary loss now adds +2.3 to +3.0 dB over MSE-only training (e.g. `blobs`: 18.9 → 21.1–21.6 dB), and the qualitative gap is larger than the numbers suggest—the MSE-only reconstructions in Figure 6 are blurred and streaked, while all three auxiliary variants recover coherent structure. Second, the central hypothesis of this paper is *not* confirmed on the PE-MLP backbone: Kac–Rice lands within 0.2–0.5 dB of FFL-with-resampling on every sampling density (`blobs` 21.1 vs. 21.6; `ramp` 22.3 vs. 22.5; `uniform` 22.5 vs. 22.7) but does not overtake it, and the normalized Sobolev loss is the strongest or tied-strongest loss throughout. On SSIM the ordering partially reverses—Kac–Rice matches FFL on `blobs` and exceeds it on `ramp` and `uniform` (0.60/0.66/0.66 vs. 0.59/0.60/0.62)—indicating the distributional loss buys structural coherence rather than pixel-wise accuracy; per-band spectra (Figure 5, right) agree, with Kac–Rice showing the lowest error of the spatial-domain losses in the top two frequency bands. But the headline claim we set out to test—a clear PSNR win for crossing densities off-grid—did not materialize.

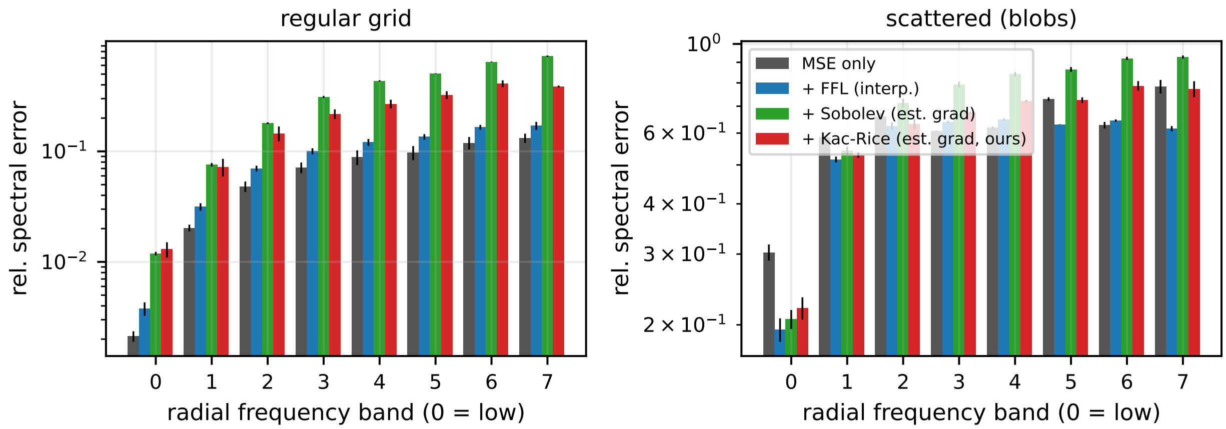

Figure 5: Per-band relative spectral error (log scale; lower is better; mean ± s.d. over 3 seeds). Left, regular grid: MSE-only is lowest in every band, mirroring Table 1. Right, scattered `blobs`: the auxiliary losses dominate MSE in the low bands, and in the two highest bands Kac–Rice attains the lowest error among the spatial-domain losses, consistent with its role as a high-frequency drive.

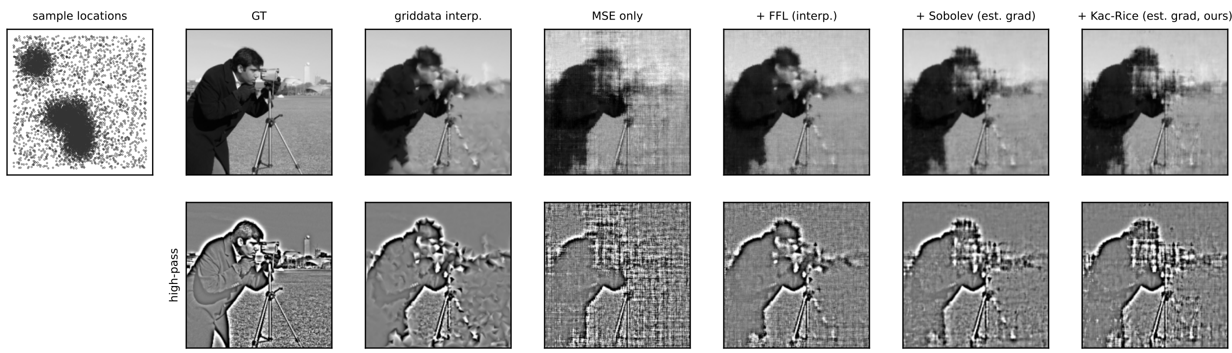

Figure 6: Qualitative results on scattered `blobs` sampling (seed 0): sample locations, reconstructions (top), high-pass residuals (bottom). MSE-only produces streaked, washed-out structure; every auxiliary loss restores coherent detail. Differences among the three auxiliary losses are subtle at this scale—consistent with the $\leq$0.5 dB spreads in Table 2.

The interpolation ceiling. The `griddata` row of Table 2 is sobering context: at this sample budget, plain linear interpolation of the scattered samples scores within noise of the *best* INR configuration on PSNR (21.8 on `blobs`). On the natural image, no loss studied here—frequency-domain or spatial—pushes a PE-MLP past the information ceiling that the interpolant itself sets (the texture task below is the sole, marginal exception). What the INR adds over `griddata` is a compact, continuous, differentiable representation, not extra pixels; auxiliary losses determine how close to the ceiling one gets.

Content dependence: crossing statistics win on texture. The synthetic multi-scale texture (Table 4) reverses the ordering. There Kac–Rice scores 27.19 ± 0.33 dB (SSIM 0.79) against FFL's 26.58 ± 0.15 (SSIM 0.74)—a +0.6 dB advantage—with Sobolev at 27.29 ± 0.13; the two spatial-domain losses match or slightly exceed the `griddata` ceiling (26.96) while FFL stays below it. The pattern is theoretically coherent: the texture is statistically homogeneous, i.e. close to the stationary-random-field regime in which crossing densities are an exact spectral functional (1), whereas `camera` is edge-dominated and non-stationary, so its crossing profile is a blunter summary. Crossing-density supervision is at its best when the signal looks like the random fields the theory was built for.

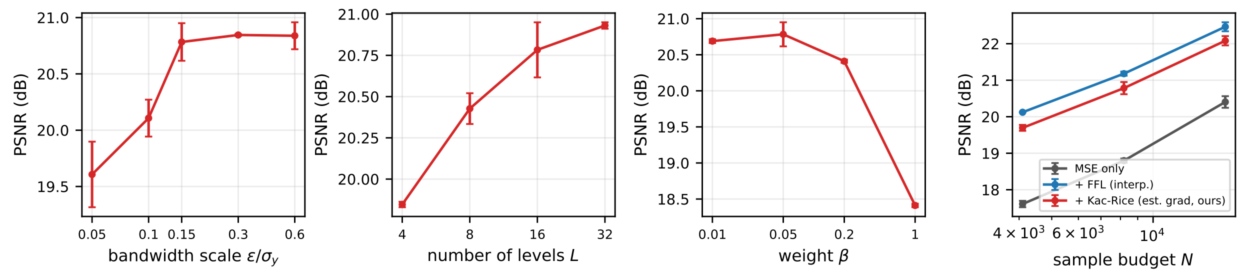

Figure 7: Ablations on `camera/blobs` (1,000 iterations, 2 seeds; error bars ± s.d.). Left to right: Dirac bandwidth scale ε/σ_y (wide plateau above 0.15), number of levels L (saturates by 16–32), loss weight β (flat over a 20× range, collapsing only at β=1), and sample budget N (orderings stable across budgets).

Oracle diagnostics: the distributional loss is robust to target smoothing. Replacing estimated gradients with oracle image gradients *helps* Sobolev (21.6 → 21.8 dB) but does not help Kac–Rice (21.1 → 20.9 dB). This is the clearest direct evidence for the robustness mechanism of Section 3.3: Sobolev consumes gradient targets pointwise and benefits from their exactness, whereas Kac–Rice only consumes their batch statistics—smoothing noise largely cancels, and sharper targets, if anything, ask for crossing mass the MSE anchor cannot yet place. The practical consequence: Kac–Rice's performance is insensitive to the quality of the gradient-estimation step, which is the step that is genuinely hard on scattered domains.

5.4 Composability with SIREN

Because the loss is orthogonal to architecture, it should stack with activation-based remedies. On the `blobs` task, SIREN+MSE scores 20.10 ± 0.20 dB (SSIM 0.56); adding Kac–Rice lifts it to 21.59 ± 0.10 (SSIM 0.65). To test whether this is specific to our loss we also ran SIREN with the two baseline auxiliaries: FFL reaches 21.92 ± 0.11 (SSIM 0.69) and Sobolev 21.50 ± 0.14 (SSIM 0.63). The honest conclusion is that *composability is a property of auxiliary spectral supervision in general*, not of crossing densities in particular—though two details favor the distributional loss here: on SIREN, Kac–Rice edges Sobolev (the reverse of their PE-MLP ordering), and every SIREN+auxiliary configuration beats its PE-MLP counterpart, so the loss and the activation fix address genuinely different parts of the problem.

5.5 Ablations

All ablations use `camera/blobs` at 1,000 iterations with 2 seeds (Figure 7); absolute numbers are therefore lower than in Table 2, and only relative sensitivity is meaningful.

Bandwidth ε. The loss is robust for $\varepsilon \geq 0.15\,\sigma_y$ (PSNR 20.8–20.9 dB across $\varepsilon/\sigma_y \in \{0.15, 0.3, 0.6\}$) and degrades below it (19.6 dB at 0.05): a too-narrow Dirac starves most batch points of gradient signal, exactly the failure mode anticipated. The practical guidance is simply "not too small"—the nuisance parameter flagged at the outset turns out to have a wide plateau rather than a narrow sweet spot.

Number of levels L. Quality improves monotonically and saturates: 19.9, 20.4, 20.8, 20.9 dB for $L = 4, 8, 16, 32$. Sixteen levels suffice.

Weight β. Performance is flat across $\beta \in [0.01, 0.2]$ (20.4–20.8 dB) and collapses only at $\beta = 1$ (18.4 dB), where the distributional term overwhelms reconstruction. The relative normalization of (5) is what makes this range transferable across signals.

Sample budget. Across $N \in \{4096, 8192, 16384\}$ the ordering is stable, with FFL ahead of Kac–Rice by a constant ∼0.4 dB (0.42/0.40/0.38). The auxiliary advantage over MSE is large at every budget but contracts slowly as sampling densifies—from +2.5/ + 2.1 dB (FFL/Kac–Rice) at N=4096 to +2.1/ + 1.7 dB at N=16384—consistent with auxiliary drives mattering most when supervision is scarcest.

6 Discussion and limitations

What we learned. Taking the experiments together, the honest verdict on the opening hypothesis is *parity with a content-dependent edge, not a general victory*: the crossing-density loss is a working, validated, mesh-free high-frequency drive that matches frequency-domain and gradient-matching supervision on edge-dominated natural-image content, beats MSE-only training by large margins wherever supervision is scarce, and overtakes FFL only where the signal is statistically homogeneous (texture)—the regime the underlying theory is actually about. Its remaining distinguishing properties are structural. First, it is the only loss in the comparison that needs neither a grid at any point in the pipeline (FFL needs one for its transform) nor trustworthy pointwise gradient targets (Sobolev's accuracy tracks target quality, while Kac–Rice is insensitive to it—the oracle diagnostic). Second, it buys structural metrics preferentially: SSIM matching (`blobs`) or exceeding (`ramp`, `uniform`) FFL's, and the best top-band spectral error among the spatial-domain losses, at a small PSNR cost. Where those structural properties are decisive—supervision without any usable grid, gradient targets too noisy to match pointwise—it is a reasonable default; where a grid or clean gradients exist, existing losses are already sufficient.

Why no win? The mechanism we can defend from the data: the crossing profile constrains L numbers per batch—a far weaker constraint than FFL's full spectrum or Sobolev's N pointwise gradients. Robustness and weakness here are two faces of the same statistic. The interpolation-ceiling observation sharpens this: at N=8,192 samples all auxiliary losses already saturate what linear interpolation extracts from the samples, so there was little headroom for a cleverer loss to express an advantage. Regimes where the ceiling itself moves—strong shape priors, composed losses, or targets (b)-style scheduled crossing profiles that inject *prior* spectral knowledge rather than re-encoding sample information—are where a distributional drive could still separate; all remain untested here.

Limitations. (i) Scope: 1D signals and 2D images only; neural SDFs from point clouds—arguably the setting where "no grid, noisy normals" bites hardest—and few-shot NeRF remain future work, and our conclusions should not be extrapolated to them. (ii) The loss constrains distributional level-set geometry, not phase: it cannot place an edge, only demand total edge mass, so it is strictly an auxiliary term. (iii) Hyper-parameters are benign but real: ε has a wide plateau above $0.15\sigma_y$, L=16 suffices, and β tolerates a 20× range—but all were tuned on the same image family used for evaluation. (iv) Double backprop makes each iteration ∼2.1× an MSE iteration, matching Sobolev. (v) Evaluation uses one natural test image plus one synthetic texture at one resolution per task; broader image suites would tighten the estimates, though the seed-level spreads are already small relative to the effects discussed. (vi) `camera` is a standard public test image; no personal or sensitive data is involved.

7 Conclusion

We turned the Rice level-crossing density into a differentiable, mesh-free, FFT-free auxiliary loss for INRs, validated the underlying estimator against exact crossing counts and the Rice formula, and evaluated it against the strongest loss-based alternatives under strict information parity. The result is a clean, honest data point for the field: crossing statistics *work* as a high-frequency drive—recovering +2.3–3.0 dB over plain reconstruction on PE-MLP and +1.4–1.8 dB on SIREN wherever supervision is scarce—but on natural images they match rather than beat frequency-domain and Sobolev supervision, pulling ahead of the frequency-domain loss only on statistically homogeneous texture, with their broader distinction being structural: no grid, no FFT, and demonstrated insensitivity to gradient-target noise. The broader point is that the geometry of level sets offers a family of spatial-domain spectral surrogates—crossing densities here; critical-point counts and Euler characteristics of excursion sets [1] are natural sequels—that remain defined exactly where frequency-domain tooling is not, and the settings that motivated this loss most strongly (point-cloud SDFs, scheduled crossing targets) remain open.

Reproducibility. Code, experiment scripts, raw result JSONs, and the estimator's unit tests are public at https://github.com/gunnerhowe/Research (directory `kac-rice`). All experiments were run on a single consumer workstation (16-core CPU / RTX 3080); the complete suite reproduces in under a day of commodity compute.

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A Hyperparameters

B Additional results

Table 3: Hyperparameters for all experiments.

Backbone (loss comparisons)	PE-MLP: ReLU, 3 hidden layers, width 256
Positional encoding	$\{\sin, \cos\}(2^k \pi x)$, $k = 0..5$ (1D), $k = 0..7$ (2D)
SIREN / FINER	width 256, 3 hidden layers, $\omega_0 = 30$; FINER default init
Optimizer	Adam, lr 10^{-3} (PE-MLP) / 5×10^{-4} (SIREN, FINER)
Schedule	cosine decay to 0.05$\times$ lr
Iterations	3,000 (1D), 2,000 (2D), 1,000 (ablations); full batch
Kac–Rice loss	$L = 16$ levels at quantiles $[0.02, 0.98]$; $\varepsilon = 0.15\sigma_y$; $\beta = 0.05$
Sobolev loss	normalized H^1; weight 0.05
FFL	$\alpha = 1$, weight 1.0; random 128^2 crops on scattered tasks
Scattered supervision	$N = 8{,}192$ samples of the 256^2 image (12.5%)
Seeds	3 (init + sampling pattern); 2 for ablations

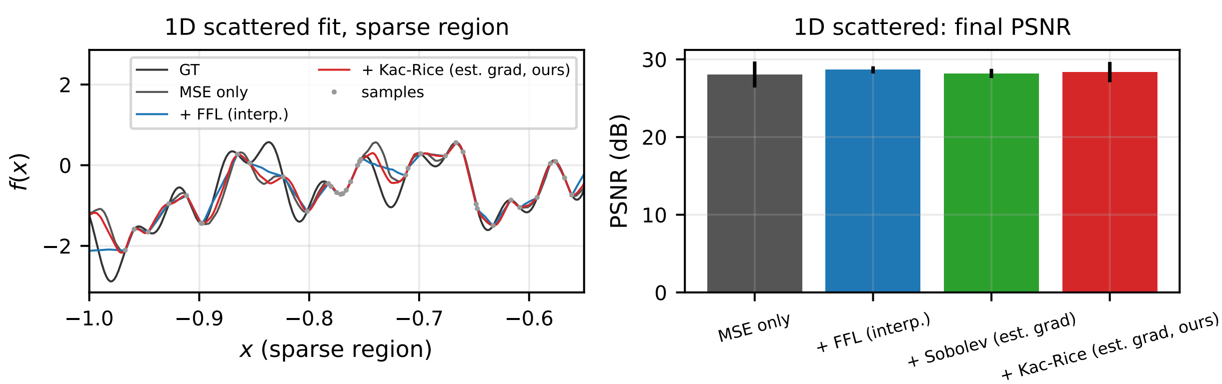

Figure 8: 1D scattered fitting (ramp density, N=384 samples, mean over 3 seeds). Left: reconstructions in the sparsely sampled region; all methods undershoot the highest tone where samples are scarce, with FFL producing the smoothest (most conservative) fit. Right: final PSNR—all four configurations tie within noise (28.1–28.6 dB). In 1D the linear interpolant is already a strong gradient/grid surrogate, and no auxiliary loss separates from the pack.

Table 4: Scattered `blobs` fitting on the synthetic multi-scale texture (256^2, N=8,192; mean $\pm$ s.d. over 3 seeds).

Config	PSNR	HF-PSNR	SSIM
MSE only	24.81 $\pm$ 0.88	19.78 $\pm$ 0.80	0.59 $\pm$ 0.06
+ FFL (interp.)	26.58 $\pm$ 0.15	21.64 $\pm$ 0.15	0.74 $\pm$ 0.01
+ Sobolev (est. grad)	27.29 $\pm$ 0.13	22.23 $\pm$ 0.22	0.81 $\pm$ 0.01
+ Kac-Rice (est. grad, ours)	27.19 $\pm$ 0.33	22.17 $\pm$ 0.10	0.79 $\pm$ 0.02